

Document Summary Assistant

Project Architecture, Technical Overview, & Technical Limitations Report

1. Executive Overview & Theme

Document Summary Assistant is an AI-powered document processing and summarization platform. It accepts PDFs and scanned document images, extracts textual content using high-speed text layers or cloud-native vision OCR, and produces clean, structured Markdown summaries paired with actionable key bullet points. The project emphasizes **zero local binary dependencies** (removing OS-level Tesseract installations), high performance via **Groq inference**, and ultimate reliability with **Google Gemini fallback**.

Key System Highlights:

- **Multi-Format Ingestion:** Supports PDFs and raster document images (.png, .jpg, .jpeg, .webp) up to 50 MB.
- **Dual AI Engine Architecture:** Primary high-speed LLM processing via Groq with seamless automatic failover to Google Gemini 3.
- **Cloud-Native Multimodal OCR:** Automatically transcribes text from scanned image PDFs and standalone image uploads without local Tesseract dependencies.
- **Interactive UX:** Dynamic summary length switching (Short, Medium, Long), single-click Markdown copying, dark/light glassmorphic theme.
- **Clean Output Pipeline:** Regex sanitization strips model reasoning tags (...) and preamble thoughts.

2. Technical Stack & Components

Layer	Technologies	Purpose / Highlights
Frontend UI	React 18, Vite, Vanilla CSS	Fast SPA with custom CSS design tokens, glassmorphism, responsive theme switching.
Backend API	FastAPI, Python 3.11+, Uvicorn	Asynchronous RESTful framework, file validation, CORS handling, provider orchestration.
Text Extraction	pdfplumber	Parses embedded PDF text layers directly with high fidelity.
Cloud OCR	Google Gemini Vision API, OpenAI Vision API	Cloud-native multimodal transcription for scanned PDFs and image files.
Primary LLM Engine	Groq API (Llama-3.3-70B, Qwen3.6-27B, GPT-OSS)	Ultra-low latency inference with dynamic model availability lookup.
Fallback LLM Engine	Google Gemini 3 (3.7-flash, 3.6-flash)	Automatic secondary provider if Groq encounters limits or API errors.
Deployment Target	Render (Backend), Vercel (Frontend)	Hosted cloud deployment setup with environment variable configuration.

3. System Architecture & Ingestion Flow

- 1. **File Reception:** Client uploads document (PDF/Image) to POST /summarize along with requested length and provider.
- 2. **Size & Type Validation:** FastAPI verifies file size is within 50MB and validates content type headers.
- 3. **Text Extraction Routing:** For PDFs, *pdfplumber* extracts selectable text layers. If no text layer exists or an image is uploaded, it routes to Cloud Vision (Gemini/OpenAI).
- 4. **LLM Summarization:** Text is dispatched to Groq API. If Groq fails or rate limits, request transparently falls back to Gemini.
- 5. **Output Sanitization:** The response passes through `_clean_summary()` regex filters before returning formatted JSON to the React frontend.

4. Technical Limitations & Constraints

Constraint / Area	Current Limitation	Impact & Details
Context Window Limit	60,000 Characters (~15,000 words)	Document text is hard-truncated at 60k chars. Large multi-chapter books are cut off rather than processed via RAG.
Stateless Architecture	No persistent database or auth	Summaries are generated in-memory. Reloading the browser page resets state. No user account management.
Supported File Formats	PDF, PNG, JPG, JPEG, WEBP	Office documents (.docx, .pptx, .xlsx) and plain markdown (.md) are not supported directly without prior conversion.
Cloud API Reliance	Third-party API key dependency	Requires active GROQ_API_KEY and GEMINI_API_KEY. Rate limits or service outages on third-party APIs affect service.
Upload Payload Cap	50 Megabytes maximum file size	Protects server memory on free tier deployments (Render free tier memory caps).